

RAPPORT DE PARCOURS RECHERCHE

Investigating focussed sets in the one-way model

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Abstract

As part of this *Parcours Recherche*, the broad field of *quantum computing* was explored, with a particular focus on *measurement-based quantum computing*. The *ZX-calculus* - a graphical language for reasoning about linear maps between qubits - was used to introduce the *one-way model*, an alternative to the traditional *quantum circuit model*. The project centered on the study of *flow* conditions, ultimately focusing on the notion of *focussed sets*. Although these sets have received limited attention in the literature, this project provided an opportunity to investigate their properties and potential interpretations. Specifically, we examined how focussed sets behave under a *local complementation* of the underlying graph and analyzed the impact of incorporating them into the correction sets of a given Pauli flow, from the preservation of the *extensiveness* of the associated correction function to two structural parameters of the flow - its *depth* and *width*.



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1 Introduction

This *Parcours Recherche* under the supervision of Dr Miriam Backens¹ was the opportunity to deep dive into the world of *quantum computing*. Motivated in the 1980s by physicists like Yuri Manin and Richard Feynman, quantum computing has emerged as a field of considerable importance due to its potential to revolutionize domains ranging from cryptography to quantum simulation.

Traditionally, quantum computations have been described in the *quantum circuit model* [1], where qubits are manipulated through sequences of quantum gates - analogous to classical logic gates - before being measured. While quantum circuits are often depicted using standard graphical notation, the *ZX-calculus* [2] has become popular as it offers a more flexible alternative. This graphical language not only simplifies circuit optimization but also provides a powerful framework for the *one-way model* [3] - an alternative model where computations are performed by adaptively measuring highly entangled graph states, as part of the *measurement-based quantum computation* (MBQC) paradigm.

A key challenge in MBQC is ensuring deterministic outcomes despite the inherent randomness of quantum measurements. This requires correction strategies governed by mathematical conditions known as *flows*. Several versions of flow [4, 5], along with related underlying notions, have been developed in the literature. Among these notions, *focussed sets* [6, 7] are sets of qubits with trivial net effect when used for corrections. Their collection has been shown to possess both group [6] and vector space [7] algebraic structures, yet their interpretation and practical role remain underexplored.

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In this project, we investigated the properties of focussed sets and their interactions with correction strategies. We studied how local complementation affects these sets and analyzed the effect of adding them to correction sets of existing Pauli flows - from preserving extensiveness of the underlying correction function to optimizing depth and width, two notions which we define.

2 Preliminaries

2.1 Quantum Information and Quantum Computation

In this first section, we provide a quick introduction to quantum information and quantum computation. In short, and to de Wolf's Lecture Notes [8], a quantum state is a *superposition* of classical states, written as a vector of amplitudes, to which we can apply either a *measurement* or a *unitary operation*.

The interested reader can find better presentations in the latest lecture notes, in Nielsen and Chuang's textbook [9], and among the many other resources listed further.

Mathematical foundations and notations

Linear algebra and other mathematics branches are fundamental to understanding quantum information and quantum computation. Below are listed some of the most important concepts.

Definition 1 A *Hilbert space* is a vector space over the complex numbers, equipped with an inner product operation.

Hilbert spaces can be finite- or infinite-dimensional.² However, quantum computations typically take place in finite-dimensional ones, of the form $\mathbb{C}^n$, and equipped with an inner product operation often denoted

$$(\Phi, \Psi) \in \mathbb{C}^n \times \mathbb{C}^n \mapsto \langle \phi | \psi \rangle = \phi^* \psi = \sum_i \phi_i^* \psi_i \in \mathbb{C}.$$

The notation $\langle \cdot | \cdot \rangle$ is called the *bra-ket notation* or *Dirac notation*, and its use is very popular in quantum mechanics and quantum computing.

For the purpose of this report, the reader needs only to know that $|\phi\rangle$ (read 'ket phi') represents a vector labeled ϕ in the considered Hilbert space H , while $\langle \phi |$ (read 'bra phi') is $|\phi\rangle^*$, that is, the adjoint of $|\phi\rangle$ or $(\phi_1^* \cdots \phi_n^*)$. It is an element of the dual space H^* , and put differently it is a linear map from H to complex numbers.

Fortunately, plugging a ket to a bra produces a bra-ket which is nothing but the inner product introduced earlier:

$$\langle \phi | \psi \rangle = |\phi\rangle^* |\psi\rangle = \phi^* \psi = \langle \phi | \psi \rangle$$

²In the latter case, the completeness of the space must be added to the definition.

The inner product operation of an Hilbert space induces a *norm* defined as follows:

$$|| \cdot || : v \mapsto ||v|| := \sqrt{\langle v|v \rangle}$$

Definition 2 A *unit vector* is a vector of norm 1.

Definition 3 Two vectors ϕ and ψ are *orthogonal* if $\langle \phi|\psi \rangle = 0$.

Definition 4 An *orthonormal basis* for an Hilbert space H is a basis for H whose vectors e_i are orthonormal, that is, they are all unit vectors and orthogonal to each other:

$$\forall i, j, \quad \langle e_i|e_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}$$

Matrices are useful to represent linear maps between vector spaces with particular bases. We assume familiarity with the matrix - linear map implicit correspondence.

Definition 5 An invertible square matrix U is *unitary* if its inverse U^{-1} equals its conjugate transpose U^* , that is, if

$$U^*U = UU^* = I$$

where I is the identity matrix.

Property 1 An $n \times n$ complex matrix U is unitary if, and only if, U is an isometry with respect to the induced norm, that is, $||Ux|| = ||x||$ for all $x \in \mathbb{C}^n$.

Definition 6 A *unitary operator* is an operator whose matrix representation is unitary.

We also assume familiarity with the tensor product (of vectors, vector spaces, linear maps) which we will denote $\otimes$, but below is reminded one of its properties.

Property 2 The tensor product is *associative*.

In particular, we have and will use that:

$$\mathbb{C}^{2m} = (\mathbb{C}^2)^{\otimes m} = \underbrace{\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \dots \otimes \mathbb{C}^2}_{m \text{ times}}$$

Quantum States

Definition 7 Let $|0\rangle, \dots, |N-1\rangle$ form an orthonormal basis of an N -dimensional Hilbert space. A *quantum state* $|\phi\rangle$ is a unit vector in this space:

$$|\phi\rangle = \alpha_0 |0\rangle + \dots + \alpha_{N-1} |N-1\rangle = \begin{pmatrix} \alpha_0 \\ \vdots \\ \alpha_{N-1} \end{pmatrix}$$

where $\alpha_0, \dots, \alpha_{N-1} \in \mathbb{C}$ and $\| |\phi\rangle \| = \sqrt{\sum_i |\alpha_i|^2} = 1$.

Considering that the $|i\rangle$ represent N different and mutually exclusive classical states of some physical system, $|\phi\rangle$ can be seen as a *superposition* of the $|i\rangle$ for which $\alpha_i \neq 0$. Intuitively, some physical system in quantum state $|\phi\rangle$ is "in *several* classical states *at the same time*", each state $|i\rangle$ having a certain *amplitude* α_i .

Quantum Bits

Most of the time, quantum information will reside in the 2-dimensional Hilbert space $\mathbb{C}^2$.³ Its standard computational basis consists of the following two vectors:

$$|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

While the fundamental unit of information in classical computation is the *bit*, which can be in one of the two states 0 and 1, the analogous concept for quantum computation is the *quantum bit*, or *qubit*, which can exist in a *superposition* of both $|0\rangle$ and $|1\rangle$.

Definition 8 A *qubit* is a unit vector in the Hilbert space $\mathbb{C}^2$, that is, a vector $|\phi\rangle$ which can be expressed as follows:

$$|\phi\rangle = \alpha |0\rangle + \beta |1\rangle \quad \text{where } \alpha, \beta \in \mathbb{C} \text{ and } |\alpha|^2 + |\beta|^2 = 1$$

Example 1 The following two states are qubits:

$$|\phi_1\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) := |+\rangle, \quad |\phi_2\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle) := |-i\rangle.$$

Physically, a qubit can be realized by any quantum system that has two distinguishable states. Examples include the spin of an electron, the polarization of a photon, or energy levels of an atom - but the physical implementation details lie way beyond the scope of this report.

³Working with units of quantum information corresponding to higher-dimensional Hilbert spaces can have interest, see *qutrits* and *qudits*.

Multiple Qubits

More generally, we can define m -qubits, which are quantum systems residing in the 2^m -dimensional tensor product space $(\mathbb{C}^2)^{\otimes m} = \mathbb{C}^{2^m}$.

From now on, we will abbreviate $|i\rangle \otimes |j\rangle$ to $|ij\rangle$. For instance, $|010\rangle$ refers to $|0\rangle \otimes |1\rangle \otimes |0\rangle$. With that notation, the standard computational basis of $\mathbb{C}^{2^m}$ is $\{|x\rangle\}_{x \in \{0,1\}^m}$.

Definition 9 An m -qubit is a unit vector in the Hilbert space $\mathbb{C}^{2^m}$, that is, a vector which can be expressed as follows:

$$|\phi\rangle = \sum_{x \in \{0,1\}^m} \alpha_x |x\rangle \quad \text{where } \forall x \in \{0,1\}^m, \alpha_x \in \mathbb{C} \text{ and } \sum_{x \in \{0,1\}^m} |\alpha_x|^2 = 1$$

Example 2 A 2-qubit and a 3-qubit:

$$|\phi_3\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |01\rangle), \quad |\phi_4\rangle = \frac{1}{\sqrt{3}}(|000\rangle + i|001\rangle + |111\rangle).$$

Entanglement

Definition 10 An m -qubit is *separable* if it can be written as the tensor product of an m_1 -qubit and an m_2 -qubit, with $m_1 + m_2 = m$ and $m_1, m_2 > 0$.

Example 3 The 2-qubit $|00\rangle$ is separable; it is $|0\rangle \otimes |0\rangle$.

It turns out that some m -qubits are not separable; that is the phenomenon of quantum *entanglement*, central to quantum theory.

Definition 11 An m -qubit that is not separable is *entangled*.

Example 4 The 2-qubit $|\Phi^+\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is entangled.

Indeed, suppose that we have some nonzero $\alpha_1, \beta_1, \alpha_2, \beta_2 \in \mathbb{C}^*$ such that

$$|\Phi^+\rangle = (\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle).$$

Expanding the right-hand side gives:

$$(\alpha_1 |0\rangle + \beta_1 |1\rangle) \otimes (\alpha_2 |0\rangle + \beta_2 |1\rangle) = \alpha_1 \alpha_2 |00\rangle + \alpha_1 \beta_2 |01\rangle + \alpha_2 \beta_1 |10\rangle + \alpha_2 \beta_2 |11\rangle$$

By identification,

$$\begin{cases} \alpha_1 \alpha_2 = \frac{1}{\sqrt{2}} \\ \alpha_1 \beta_2 = 0 \\ \beta_1 \alpha_2 = 0 \\ \beta_1 \beta_2 = \frac{1}{\sqrt{2}} \end{cases}$$

Since the coefficients were assumed nonzero, the second and third equations exhibit a contradiction. Therefore, $|\Phi^+\rangle$ is not separable.

The $|\Phi^+\rangle$ state is called an EPR pair in honor of Einstein, Podolsky, and Rosen [10].

Quantum Measurement

Quantum mechanics teaches us that quantum measurement is fundamentally probabilistic. Moreover, it can disturb the system.

Definition 12 When measuring $|\phi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle$, one will get outcome $|x_0\rangle$ as a result with probability $|\alpha_{x_0}|^2$. If so, $|\phi\rangle$ “collapses” to $|x_0\rangle$.

The above measurement occurs in the standard computational basis. Measurements in other bases are possible, and even fundamental in the Measurement-Based Quantum Computation model we will meet further.

Unitary evolution

Instead of measuring a quantum state, one might also want to apply some operation to it. Quantum mechanics only allows unitaries.

Property 3 Only unitary operators can be applied to quantum states.

It follows that any (non-measuring) operation on quantum states is reversible: one can always apply U^{-1} , the inverse of the unitary U , to “undo” the action of U . On the other hand, measurements are clearly non-reversible.

Elementary Gates

In analogy to classical logic gates (AND, OR, NOT, etc.), unitaries can be viewed as quantum logic gates.⁴

Definition 13 A *quantum logic gate*, or simply *gate*, is a unitary operator.

Note that unlike many classical logic gates, quantum ones are - since they are unitaries - always reversible.

There exists an uncountably infinite number of gates. Below are listed some of the most popular. As by a precedent remark, we only give their matrix representation. The reader is free to verify that they are indeed unitary matrices.

Example 5 The *Pauli matrices* are examples of 1-qubit gates:

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

For instance, the *bitflip gate* X can be seen as a NOT gate since it swaps $|0\rangle$ and $|1\rangle$.

⁴Some authors require that these unitaries act on only one or two qubits, others on “a small number” of qubits, but that has no importance to us.

Example 6 Another important 1-qubit gate is the *phase gate* R_ϕ , which rotates the phase of the $|1\rangle$ -state by an angle ϕ :

$$R_\phi = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}.$$

Example 7 The most important 1-qubit gate might be the *Hadamard gate*, represented by:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Hadamard gates are used to produce equal superposition states from computational basis states:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) := |+\rangle, \quad H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) := |-\rangle.$$

Moreover, $H^* = H$. That is, the Hadamard gate is Hermitian. Since it is also unitary, it is finally self-inverse:

$$HH = I.$$

Example 8 An example of a 2-qubit gate is the *controlled-NOT gate* CNOT. It negates the second bit (*target* bit) of its input if the first bit (*control* bit) is 1, and does nothing otherwise:

$$\text{CNOT}|1\rangle|b\rangle = |1\rangle|1-b\rangle, \quad \text{CNOT}|0\rangle|b\rangle = |0\rangle|b\rangle.$$

In matrix form, this is:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Example 9 More generally, if U is some n -qubit unitary matrix, then the *controlled- U gate* corresponds to the following $2^{n+1} \times 2^{n+1}$ unitary matrix:

$$CU = \begin{pmatrix} I_{2^n} & 0_{2^n} \\ 0_{2^n} & U \end{pmatrix}.$$

Quantum Circuits

The most popular quantum computation model is the *quantum circuit model*, first introduced in [1] and which we present here.

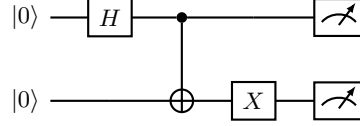
Definition 14 A *quantum circuit* is a large unitary map constructed from the composition (sequential application) and tensor product (parallel application) of a (finite) set of elementary gates acting on one or more qubits.

In the quantum circuit model, a computation proceeds in three steps:

1. Prepare N qubits in a known initial state, typically $|0\rangle^{\otimes N}$.
2. Apply a quantum circuit to the qubits.
3. Measure one or more of the qubits.

As in classical circuit theory, quantum computations can be represented diagrammatically. In such circuit diagrams, qubits are represented by lines going from left to right, and quantum gates are represented by boxes.

Example 10 Below is the diagrammatic representation of an example computation in the quantum circuit model. The middle vertical line corresponds to a CNOT gate, and the rightmost boxes represent measurements.



(Approximately) universal gate sets

When constructing quantum circuits, one will often restrict to some well-defined set of elementary gates.

Definition 15 A set of gates $\mathcal{G}$ is *universal* if the gates in $\mathcal{G}$ can be combined to construct any N -qubit unitary U .

Example 11 The set $\mathcal{G} = \{\text{CNOT}, H\} \cup \{R_\alpha, \alpha \in [0, 2\pi[\}$ is universal.

Since there are uncountably many N -qubit unitaries as by a precedent remark, universal gate sets are necessarily infinite. As a result, finite *approximately* universal gate sets are also considered. Their elements allow to approximate any N -qubit unitary up to arbitrarily high precision.

Definition 16 For some $\varepsilon > 0$, a unitary U' ε -approximates U if for all quantum state $|\phi\rangle$,

$$\|U|\phi\rangle - U'|\phi\rangle\| \leq \varepsilon.$$

Definition 17 A set of gates $\mathcal{G}$ is *approximately universal* if for any $\varepsilon > 0$, any unitary can be ε -approximated by a unitary constructed by the combination of gates from $\mathcal{G}$.

Example 12 ([9]) The following finite set is approximately universal:

$$\text{Clifford+T} := \{\text{CNOT}, H\} \cup \{T := R_{\pi/4}\}$$

Quantum Algorithms

The quantum circuit model, together with approximately universal gate sets, allows us to describe a wide range of quantum algorithms that have had a significant impact in both theory and practice. Examples include Grover's algorithm [11] and Shor's algorithm [12]. The interested reader may refer to [13, 2.4] for a better description.

2.2 The ZX-calculus

We provide a brief overview over the ZX-calculus, a diagrammatic language similar to the quantum circuit notation seen before and introduced by Bob Coecke and Ross Duncan in [2]. This section is much inspired by [14, 2.1], and we refer to [13, Chapter 3] for a more detailed introduction.

In short, ZX-diagrams consist of green and red nodes called *spiders* which are connected together by *wires*.

ZX-diagrams

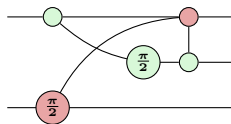
Spiders are linear maps which can have any number of input or output wires. There are two varieties: Z spiders, depicted in green, and X spiders, depicted in red :

$$\begin{aligned}
\text{(Z)} \quad & \text{Diagram: A green circle with } \alpha \text{ inside, four lines entering from the top and four from the bottom.} & := |0 \cdots 0\rangle \langle 0 \cdots 0| + e^{i\alpha} |1 \cdots 1\rangle \langle 1 \cdots 1| \\
\text{(X)} \quad & \text{Diagram: A red circle with } \alpha \text{ inside, four lines entering from the top and four from the bottom.} & := |+\cdots+\rangle \langle +\cdots+| + e^{i\alpha} |-\cdots-\rangle \langle -\cdots-|
\end{aligned}$$

One might omit writing the α angle on a spider, in which case that angle is assumed to be 0 and the spider is called *phase-free*.

A ZX-diagram with m input wires and n output wires then represents a linear map $(\mathbb{C}^2)^{\otimes m} \mapsto (\mathbb{C}^2)^{\otimes n}$ built by the composition and tensor product of the latest linear maps. Remark that diagrams with no input and n outputs represent vectors in $(\mathbb{C}^2)^{\otimes n}$, i.e. n -qubits.

Example 13 An example ZX-diagram, with 2 input wires and 3 output wires hence representing a linear map from $(\mathbb{C}^2)^{\otimes 2}$ to $(\mathbb{C}^2)^{\otimes 3}$.



Example 14 Another ZX-diagram, this time with no input and 3 outputs. Note this represents the vector $|000\rangle + |111\rangle$, that is, an unnormalized 3-qubit state often called the GHZ state.

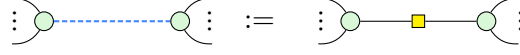


Two ZX-diagrams can be composed by joining the output wires of the first to the input wires of the second. One can also form their tensor product by simply stacking the two diagrams vertically.

Because its use is very frequent, note that a specific symbol - a yellow square - is often introduced for the Hadamard gate:



When placed between two spiders, it will be replaced by a blue dashed edge:



The rules of the ZX-calculus

In order to obtain what we call the *ZX-calculus*, ZX-diagrams must be equipped with a set of rewrite rules. There exist different popular sets of rewrite rules, each of which corresponding to a different version of the ZX-calculus. Here we only present the one adopted in [14].

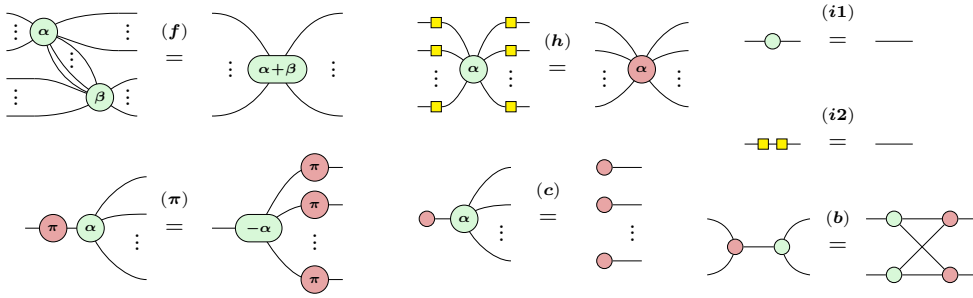
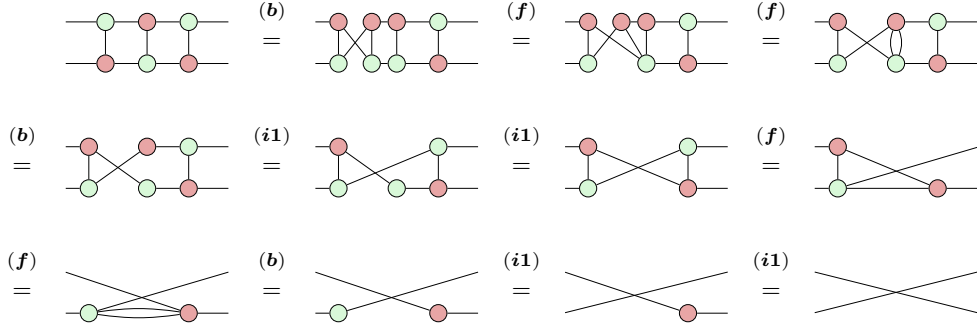


Fig. 1 A convenient presentation for the ZX-calculus. These rules hold for all $\alpha, \beta \in [0, 2\pi)$, and due to (h) and (i2) all rules also hold with the colours interchanged. Note the ellipsis should be read as ‘0 or more’, hence the spiders on the LHS of (f) are connected by one or more wires.

In the version presented above, equalities are preserved up to a global scalar.

A rule that is however common to all versions of the ZX-calculus is that *only connectivity matters*. That is, the interpretation of a ZX-diagram is invariant under moving the vertices around in the plane, bending, unbending, crossing, and uncrossing wires, so long as the connectivity and the order of the inputs and outputs is maintained.

Example 15 In this example - inspired by the zxcalculus.com website landing page - we use some of the rules presented above in order to derive a simple swap gate from three CNOTs.



Some properties

The ZX-calculus is *universal*, in the sense that any linear map between qubits can be expressed as a ZX-diagram [13].

When restricted to some families (or *fragments*) of linear maps, it can also be made *complete*, in the sense that given two non-zero ZX-diagrams that represent the same linear map, there exists a series of rewrites transforming one into the other. If a meta rule that allows non-zero scalar subdiagrams to be dropped is added, the version presented above is complete when restricted to *Clifford* ZX-diagrams, ie. diagrams whose phases are all multiples of $\pi/2$ [15, 16]. Extensions to the ZX-calculus which are complete for the larger *Clifford+T* family [17] or even for *all* ZX-diagrams [18] have been introduced in the latest decade.

There and back again

As mentioned above, the ZX-calculus is universal in the sense that it can represent any linear map. In particular, it can represent any unitary map, that is, any quantum circuit. And given a quantum circuit constructed from - say - the approximately universal *Clifford+T* gate set, translating it into a ZX-diagram is actually straightforward, with the help of a translation table.

However, the converse operation turns out to be more complicated. In fact, the problem of translating a ZX-diagram into a quantum circuit - known as the *circuit extraction problem* [19] - is not solvable in the general case. Indeed, it is clear that there exist maps that are linear but not unitary. But even when restricting to ZX-diagrams representing unitary maps, the problem is known to be #P-hard [20]. Nevertheless, it turns out to be efficiently solvable for specific classes of ZX-diagrams that possess certain structural properties. For instance, Backens et al. [14] provide an efficient algorithm for circuit extraction from ZX-diagrams which represent measurement patterns with extended gflow - a notion which will be made clearer later in this report.

Back to quantum measurements

In Definition 12, we defined the quantum measurement of an n -qubit state $|\phi\rangle$ as its projection onto the standard computational basis $\{|x\rangle\}_{x \in \{0,1\}^n}$.

In fact, one may choose to measure only a single qubit of such a state. Furthermore, the measurement need not be performed in the computational basis: any orthonormal basis of $\mathbb{C}^2$ may be used to define the measurement outcomes for that qubit.

The following bases of $\mathbb{C}^2$ - called the *Pauli bases* - are the most common ones:

$$\mathcal{B}_Z = \{|0\rangle, |1\rangle\}, \mathcal{B}_Y = \{|+i\rangle, |-i\rangle\}, \mathcal{B}_X = \{|+\rangle, |-\rangle\}$$

where $|\pm i\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm i|1\rangle)$ and $|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$.

We will also consider the following:

$$\mathcal{B}_{XY,\alpha} = \{|+_{XY,\alpha}\rangle, |-_{XY,\alpha}\rangle\}, \mathcal{B}_{XZ,\alpha} = \{|+_{XZ,\alpha}\rangle, |-_{XZ,\alpha}\rangle\}, \mathcal{B}_{YZ,\alpha} = \{|+_{YZ,\alpha}\rangle, |-_{YZ,\alpha}\rangle\},$$

where $\alpha \in [0, 2\pi[$ and:

$$\begin{aligned} |+_{XY,\alpha}\rangle &:= |0\rangle + e^{i\alpha} |1\rangle, & |-_{XY,\alpha}\rangle &:= |0\rangle - e^{i\alpha} |1\rangle, \\ |+_{XZ,\alpha}\rangle &:= |+i\rangle + e^{i\alpha} |-i\rangle, & |-_{XZ,\alpha}\rangle &:= |+i\rangle - e^{i\alpha} |-i\rangle, \\ |+_{YZ,\alpha}\rangle &:= |+\rangle + e^{i\alpha} |-\rangle, & |-_{YZ,\alpha}\rangle &:= |+\rangle - e^{i\alpha} |-\rangle. \end{aligned}$$

Now, how are single-qubit measurements represented in the ZX ? Let us say that we are measuring the i -th qubit of a quantum state $|\phi\rangle$ in the basis $\mathcal{B}_{XY,\alpha}$. That is, we are projecting the i -th qubit of $|\phi\rangle$ onto the $\{|+_{XY,\alpha}\rangle, |-_{XY,\alpha}\rangle\}$ basis.

- Suppose that we obtain the $|+_{XY,\alpha}\rangle$ state. In the Dirac notation, this corresponds to applying $\langle +_{XY,\alpha}| = \langle 0| + e^{-i\alpha} \langle 1|$ to that single qubit. And in the ZX-calculus, that is simply represented by connecting a Z spider with an α phase ⁵ to that same qubit: 
- Now if we instead obtain the $|-_{XY,\alpha}\rangle$ state, this corresponds to applying $\langle -_{XY,\alpha}| = \langle 0| - e^{-i\alpha} \langle 1| = \langle 0| + e^{i(-\alpha+\pi)} \langle 1|$, and is represented in the ZX by connecting a Z spider with an $\alpha + \pi$ phase ⁶: 

We can give a unified ZX representation by introducing a boolean variable b representing the outcome of the measurement: $b = 0$ if we measure $|+_{XY,\alpha}\rangle$, $b = 1$ if we measure $|-_{XY,\alpha}\rangle$. With that, a single-qubit measurement in the basis $\mathcal{B}_{XY,\alpha}$ is simply represented by connecting a Z spider with an $\alpha + b\pi$ phase:

$$\text{---} \langle \alpha + b\pi$$

With an analogous reasoning for all other orthonormal bases presented above, we finally obtain Table 1.

See [13, 2.2.4] for more details on measurements, especially on their visualization in the widely-known Bloch sphere.

⁵ Actually, an $-\alpha$ phase, but the minus sign is often dropped for simplicity here.

⁶ Same remark as Footnote 5

Measurement basis	$\mathcal{B}_{XY,\alpha}$	$\mathcal{B}_{XZ,\alpha}$	$\mathcal{B}_{YZ,\alpha}$	$\mathcal{B}_Z$	$\mathcal{B}_Y$	$\mathcal{B}_X$
ZX-diagram						

Table 1 Translation of measurement effects in the ZX for different bases.

2.3 Measurement-Based Quantum Computation

The quantum circuit model presented earlier is not the only way in which one can realize a quantum computation. Introduced in [3], the *one-way model* is a promising instance of the *Measurement-Based Quantum Computation* (MBQC) paradigm. In the one-way model, a computation proceeds as follows:

1. Prepare some highly entangled quantum state, called a *graph state*.
2. Gradually consume the graph state by applying single qubit measurements.

While the ‘actual computation’ happens through the application of gates in the quantum circuit model, we can say that it only happens through the application of measurements in the one-way model.

Since quantum measurements are fundamentally probabilistic, a computation in the one-way model requires corrections in order to be deterministic ‘overall’. In that goal, there must exist some correction strategy - called a *flow* - which conditions how the basis of certain measurements should be adapted upon the outcome of others and introduces a measurement order.

Unfortunately, not all labelled open graphs - the objects which describe the fundamental properties of a computation in the one-way model, defined below - have flow, ie. can be used for deterministic computation. Different versions of flow which guarantee different levels of determinism have been developed in the literature, but in this report we will solely focus on Pauli flow which presence is necessary and sufficient for ‘robust’ determinism.

Note that the one-way model is universal, in the sense that it can implement any unitary. Indeed, one can for example implement all gates from the universal *Clifford+T* gate set, as proven in [3].

Finally, the ZX-calculus presented earlier turns out be particularly well adapted for describing computations in the one-way model, so we will make use of it.⁷

Graph state

As introduced earlier, the basic resource of a computation in the one-way model is a graph state.

⁷Actually, the development of the ZX-calculus was originally motivated by the desire to describe MBQC in a better way.

Definition 18 (Graph state) Given an undirected graph $G = (V, E)$, the *graph state* associated with G is the quantum state $|G\rangle$ defined as follows:

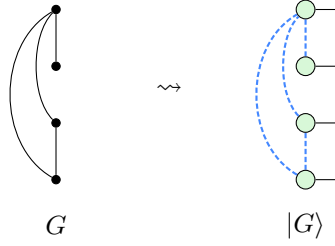
$$|G\rangle := \prod_{(v,w) \in E} CZ_{v,w} |+\rangle^{\otimes |V|}$$

where $CZ_{v,w}$ is a Controlled-Z gate - that is, an entangling operator - between qubits v and w .

Remark 1 In other words, given G , preparing $|G\rangle$ consists of:

- preparing a qubit in the state $|+\rangle$ ( in the ZX) for each vertex of G , and
- applying an entangling operator () between the corresponding qubits of all edges of G .

Example 16 Below is an example mathematical graph G and the representation of its associated graph state $|G\rangle$ in the ZX.

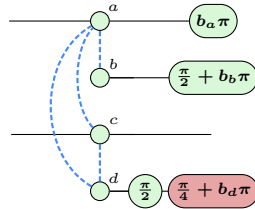


MBQC in the ZX

In order to represent a computation in the one-way model with the ZX-calculus, we need to define a new family of ZX-diagrams.

Definition 19 (MBQC-form ZX-diagram) A ZX-diagram is in *MBQC-form* if it consists of a graph state where we have added input wires or measurement effects to some vertices.

Example 17 Below is an example MBQC-form ZX-diagram whose underlying graph comes from Example 16.

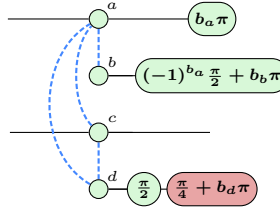


Recall the representation of single-qubit measurements in the ZX-calculus has been introduced in 2.2. From that, the diagram should be read as follows: qubit associated with vertex a ('qubit a ') is an input and is measured in the Pauli basis $\mathcal{B}_Z$, qubit b is measured in the Pauli basis $\mathcal{B}_Y$, qubit c is both an input and an output, qubit d is measured in the rotated Pauli basis $\mathcal{B}_{XZ, \pi/4}$, and b_a, b_b and b_d are boolean variables representing measurement outcomes.

Now suppose the ZX-diagram in the previous example represents a computation in which the three measurement outcomes are the output variables. Since quantum measurements are inherently probabilistic, this computation is not deterministic: the b_i variables are random.

In order to achieve overall determinism, we must somehow 'get rid' of these variables. To do this, we can adapt the measurement angles based on the outcomes of previous measurements. In other words, if we define a measurement order - that is, a sequence in which the single-qubit measurements are performed - we can modify the measurement basis of the i -th qubit based on the known outcomes of earlier measurements.

Example 18 If we say that qubit b will be measured after qubit a in our previous example, then we can adapt the basis in which b will be measured to the boolean b_a which represents the outcome of the measurement of a .



From this point and with the help of the rewriting rules of the ZX-calculus, we can show that it is sometimes possible to push all undesired measurement outcomes outside of such a ZX-diagram - in a process called *feed-forward* - so that the computation becomes deterministic up to Pauli errors at the end which we know how to deal with.

See [13, 8.1] for more details. In such a more detailed introduction to correction strategies appears the notion of odd neighborhood, which we present now.

(Closed) odd neighborhood

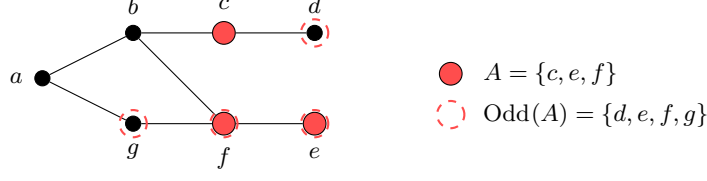
Definition 20 Let $G = (V, E)$ be a graph $A \subseteq V$ some subset of its vertices. We define the *odd neighborhood* of A in G as :

$$\text{Odd}_G(A) := \{w \in V, |N_G(w) \cap A| \equiv 1 \pmod{2}\}$$

where $N(w)$ denotes the regular neighborhood of a vertex w , ie. its set of neighbors.

Put differently, the odd neighborhood of a set of vertices A is the set of vertices $w \in V$ which are adjacent to an odd number of elements of A .

Example 19 Below is an illustration of the odd neighborhood.



For instance, $g \in \text{Odd}(A)$ because g is adjacent to exactly one (odd) element of A , while $b \notin \text{Odd}(A)$ because b is adjacent to exactly two (even) elements of A .

Remark 2 We also have $\text{Odd}_G(A) = \Delta_{v \in A} N_G(v)$.

Proof That can be proved by induction on the size of A .

For $A = \emptyset$ of size null, that is immediate: $\text{Odd}_G(\emptyset) = \emptyset = \Delta_{v \in \emptyset} N_G(v)$.

Let $n > 0$ and $A \subseteq V$ of size $n + 1$. Also let $w \in A$. We have the following equalities:

$$\begin{aligned}
\Delta_{v \in A} N_G(v) &= (\Delta_{v \in A \setminus \{w\}} N_G(v)) \Delta N_G(w) \\
&= \text{Odd}(A \setminus \{w\}) \Delta N_G(w) \text{ by induction, } A \setminus \{w\} \text{ being of size } n \\
&= \{v \in V, |N_G(v) \cap (A \setminus \{w\})| \equiv 1 \pmod{2}\} \Delta N_G(w) \\
&= (\{v \in V, |N_G(v) \cap A| \equiv 1 \pmod{2} \text{ and } w \notin N_G(v)\} \\
&\quad \cup \{v \in V, |N_G(v) \cap A| \equiv 0 \pmod{2} \text{ and } w \in N_G(v)\}) \Delta N_G(w) \\
&= (\{v \in V, |N_G(v) \cap A| \equiv 1 \pmod{2} \text{ and } v \notin N_G(w)\} \\
&\quad \cup \{v \in V, |N_G(v) \cap A| \equiv 0 \pmod{2} \text{ and } v \in N_G(w)\}) \Delta N_G(w) \text{ because } w \in N_G(v) \iff v \in N_G(w) \\
&= ((\text{Odd}(A) \cap \overline{N_G(w)}) \cup (\overline{\text{Odd}(A)} \cap N_G(w))) \Delta N_G(w) \\
&= (\text{Odd}(A) \Delta N_G(w)) \Delta N_G(w) \\
&= \text{Odd}(A) \Delta N_G(w) \Delta N_G(w) \\
&= \text{Odd}(A) \Delta \emptyset \\
&= \text{Odd}(A)
\end{aligned}$$

□

Lemma 1 Let $A, B \subseteq V$ be two sets of vertices of $G = (V, E)$. We have the following equality:

$$\text{Odd}_G(A \Delta B) = \text{Odd}_G(A) \Delta \text{Odd}_G(B)$$

Proof That is immediate with the latest remark:

$$\text{Odd}_G(A \Delta B) = \Delta_{v \in A \Delta B} N_G(v) = (\Delta_{v \in A} N_G(v)) \Delta (\Delta_{v \in B} N_G(v)) = \text{Odd}_G(A) \Delta \text{Odd}_G(B)$$

□

Definition 21 The *closed odd neighborhood* is defined as :

$$\text{Odd}_G[A] := \text{Odd}_G(A) \Delta A$$

Example 20 In Example 19, $\text{Odd}_G\{A\} = \{a, e, f, d\} \Delta \{e, f\} = \{a, d\}$.

Lemma 2 Let $A, B \subseteq V$ be two sets of vertices of $G = (V, E)$. We have the following equality:

$$\text{Odd}_G\{A \Delta B\} = \text{Odd}_G\{A\} \Delta \text{Odd}_G\{B\}$$

Proof We use the latest result and the associativity of Δ .

$$\begin{aligned} \text{Odd}_G\{A \Delta B\} &= \text{Odd}_G(A \Delta B) \Delta (A \Delta B) \\ &= (\text{Odd}_G(A) \Delta \text{Odd}_G(B)) \Delta (A \Delta B) \\ &= (\text{Odd}_G(A) \Delta A) \Delta (\text{Odd}_G(B) \Delta B) \\ &= \text{Odd}_G\{A\} \Delta \text{Odd}_G\{B\} \end{aligned}$$

□

MBQC formalization

With that notion and the hint of intuition given above, we can now better formalize what computations in the one-way model actually are. First of all, the underlying graph and the input and output vertices are captured by the notion of open graph.

Definition 22 (Open graph) An *open graph* is a tuple (G, I, O) where $G = (V, E)$ is an undirected graph and $I, O \subseteq V$ are respectively called the sets of *input* and *output* vertices.

Remark 3 We will often use the following notations:

- $\bar{O} := V \setminus O$ is the set of non-output vertices,
- $\bar{I} := V \setminus I$ is the set of non-input vertices,
- $n := |V|$ is the total number of vertices
- n_I is the number of inputs and
- n_O is the number of outputs.

Then, we need to formalise the measurement of non-output vertices. That is done through the notion of measurement labelling, which finally brings us to the object of labelled open graph.

Definition 23 (Measurement labelling) A *measurement labelling* for an open graph (G, I, O) is a function

$$\lambda : \bar{O} \mapsto \{X, Y, Z, XY, XZ, YZ\}$$

satisfying $\forall i \in I \setminus O, \lambda(i) \in \{X, Y, XY\}$.

Definition 24 (Labelled open graph) A *labelled open graph* is a tuple $\Gamma = (G, I, O, \lambda)$ where (G, I, O) is an open graph and λ is a measurement labelling for (G, I, O) .

Now we have to capture the correction strategies mentioned above. They first require us to define corrections functions.

Definition 25 (Correction function) Let $\Gamma = (G, I, O, \lambda)$ be a labelled open graph. A *correction function* for Γ is a function $c : \bar{O} \mapsto \mathcal{P}(\bar{I})$.

Remark 4 Given our precedent ZX-intuition, a correction function c can be interpreted as follows: for each vertex v , the set $c(v)$ defines the set of vertices to which one must apply X in case of the undesired outcome, with the set of vertices requiring a Z given by its odd neighborhood $\text{Odd}(c(v))$.

Along with a partial order determining the order according to which vertices must be measured for the corrections to be possible, this finally brings us to the notion of Pauli flow.

Definition 26 (Pauli flow, adapted from Definition 5, [5]) A *Pauli flow* for a labelled open graph $\Gamma = (G, I, O, \lambda)$ is a pair $(c, \prec)$, where c is a correction function for Γ and $\prec$ is a strict partial order on $\bar{O}$ such that for all $u \in \bar{O}$:

- (P1) $\forall v \in c(u), u \neq v \wedge \lambda(v) \notin \{X, Y\} \implies u \prec v$
- (P2) $\forall v \in \text{Odd}(c(u)), u \neq v \wedge \lambda(v) \notin \{Y, Z\} \implies u \prec v$
- (P3) $\forall v \in \bar{O}, \neg(u \prec v) \wedge u \neq v \wedge \lambda(v) = Y \implies v \notin \text{Odd}[c(u)]$
- (P4) $\lambda(u) = XY \implies u \notin c(u) \wedge u \in \text{Odd}(c(u))$
- (P5) $\lambda(u) = XZ \implies u \in c(u) \wedge u \in \text{Odd}(c(u))$
- (P6) $\lambda(u) = YZ \implies u \in c(u) \wedge u \notin \text{Odd}(c(u))$
- (P7) $\lambda(u) = X \implies u \in \text{Odd}(c(u))$
- (P8) $\lambda(u) = Z \implies u \in c(u)$
- (P9) $\lambda(u) = Y \implies u \in \text{Odd}[c(u)]$

The sets $c(v)$ for $v \in \bar{O}$ are called the *correction sets*.

As announced earlier, the presence of a Pauli flow is necessary and sufficient for ‘robust’ (or ‘strong uniform stepwise’) determinism. See [5] for more details, but among other things ‘robust’ means that determinism holds for all measurement angles values.⁸

Theorem 1 (Theorems 2.39, 2.40, [14]) *A one-way computation is robustly deterministic if and only if its associated labelled open graph has Pauli flow.*

Remark 5 As mentioned, other types of flow were developed in the literature [4, 5]. For example, when restricting to XY measurements, the causal flow from Danos and Kashefi [4] requires less strict conditions. However, it is sufficient but not necessary for robust determinism.

⁸That is why they are not part of the definition of a labelled open graph.

Definition 27 (Focussed Pauli flow, [7]) The Pauli flow $(c, \prec)$ is *focussed* when for all $v \in \bar{O}$ the following hold:

- (FPX) $\forall w \in (\bar{O} \setminus \{v\}) \cap c(v), \quad \lambda(w) \in \{XY, X, Y\},$
- (FPZ) $\forall w \in (\bar{O} \setminus \{v\}) \cap \text{Odd}(c(v)), \quad \lambda(w) \in \{XZ, YZ, Y, Z\},$ and
- (FPY) $\forall w \in (\bar{O} \setminus \{v\}), \quad \lambda(w) = Y \implies w \notin \text{Odd}[\mathcal{A}].$

Theorem 2 (Lemma 4.6, [6]) *For any labelled open graph Γ , if a Pauli flow exists then there also exists a focussed Pauli flow.*

For the next few definitions, let Γ be a labelled open graph and let c be a correction function for Γ .

Definition 28 (Focussed correction function, Definition 2.7, [7]) We say that c is *focussed* when it satisfies (FPX), (FPZ) and (FPY).

Definition 29 (Minimal order, Definition 2.8, [7]) We call $\prec$ *minimal* when no proper subset of $\prec$ forms a Pauli flow with c .

Definition 30 (Extensive correction function, Definition 2.9, [7]) We call c (*focussed*) *extensive* when there exists $\prec$ such that $(c, \prec)$ is a (focussed) Pauli flow for Γ .

Definition 31 (Induced relation, Definition 2.10, [7]) The *induced relation* $\triangleleft_c$ is the minimal relation on $\bar{O}$ implied by (P1), (P2) and (P3). That is, $u \triangleleft_c v$ if and only if at least one of the following holds:

- $\forall v \in c(u), u \neq v \wedge \lambda(v) \notin \{X, Y\}$ (P1)
- $\forall v \in \text{Odd}(c(u)), u \neq v \wedge \lambda(v) \notin \{Y, Z\}$ (P2)
- $v \in \text{Odd}[c(u)] \wedge u \neq v \wedge \lambda(v) = Y$ (P3)

We denote the transitive closure⁹ of $\triangleleft_c$ as $\prec_c$.

Lemma 3 (Minimal order containment, Lemma 2.11, [7]) Let $\prec$ be an order such that $(c, \prec)$ is a Pauli flow for Γ . Then $\prec_c \subseteq \prec$.

Theorem 3 (Extending a correction function to a Pauli flow, Theorem 2.12, [7]) *The correction function c is (focussed) extensive if, and only if, $(c, \prec_c)$ is a minimal (focussed) Pauli flow for Γ .*

Lemma 4 (All Paulis are initial, Lemma 2.13, [7]) We have that :

$$\forall u \in \bar{O}, \lambda(u) \in \{X, Y, Z\} \implies \forall v \in \bar{O}, \neg(v \prec_c u).$$

⁹That is, the smallest relation that contains the original order and is transitive.

2.4 Focussed sets

To our knowledge, only two references to the notion of focussed (or focused) sets were made in the literature. Simmons first introduced them in [6], while Mitosek and Backens further explored their properties in [7].

Definition 32 (Focussed set, [7, Definition 3.21]) Given a labelled open graph $\Gamma = (G, I, O, \lambda)$, a set $\mathcal{A} \subseteq \bar{I}$ is *focussed over* $S \subseteq \bar{O}$ if:

- (FX) $\forall w \in S \cap \mathcal{A}, \quad \lambda(w) \in \{XY, X, Y\},$
- (FZ) $\forall w \in S \cap \text{Odd}(\mathcal{A}), \quad \lambda(w) \in \{XZ, YZ, Y, Z\},$ and
- (FY) $\forall w \in S, \quad \lambda(w) = Y \implies w \notin \text{Odd}[\mathcal{A}].$

A set $\mathcal{A} \subseteq \bar{I}$ is simply called *focussed* if it is focussed over $\bar{O}$. The collection of all focussed sets for Γ is denoted $\mathfrak{F}_\Gamma$.

Remark 6 Put in parallel with Definition 27, this new definition says that the Pauli flow $(c, \prec)$ is focussed if and only if the set $c(v)$ is focussed over $\bar{O} \setminus \{v\}$ for all $v \in \bar{O}$.

To Mitosek and Backens [7], focussed sets are *sets with trivial net effect when used for corrections*. Put differently, we can say that the Pauli X and Z corrections arising from the measurements of the vertices within a focussed set cancel each other out when applied to other vertices.

$\mathfrak{F}_\Gamma$ as a group

Simmons enumerates a few results on focussed sets and Pauli flows in [6], ultimately showing that one can always reach the focussing conditions from any Pauli flow. As a corollary, he shows that the focussed sets of a labelled open graph form a group.

Property 4 ([6, Lemma B.8]) The collection of focussed sets $\mathfrak{F}_\Gamma$ has a group structure under the symmetric difference operator Δ .

Proof That can directly be proved from the latest definition of focussed sets:

- (closure) Let $\mathcal{A}, \mathcal{B}$ be two focussed sets of Γ . From the precedent lemmas, we have $\text{Odd}(\mathcal{A} \Delta \mathcal{B}) = \text{Odd}(\mathcal{A}) \Delta \text{Odd}(\mathcal{B})$ and $\text{Odd}[\mathcal{A} \Delta \mathcal{B}] = \text{Odd}[\mathcal{A}] \Delta \text{Odd}[\mathcal{B}]$. With that and the definition of Δ in mind, all three conditions (FX), (FY) and (FZ) are straightforward to verify.
- (associativity) The symmetric difference Δ is associative.
- (identity) The empty set $\emptyset$ is a focussed set and it is such that $\forall \mathcal{A} \in \mathfrak{F}_\Gamma, \mathcal{A} \Delta \emptyset = \emptyset \Delta \mathcal{A} = \mathcal{A}$.
- (inverses) Each focussed set is self-inverse: $\forall \mathcal{A} \in \mathfrak{F}_\Gamma, \mathcal{A} \Delta \mathcal{A} = \emptyset$.

□

Simmons also studied the size of that group and showed that it is exponential in the difference between the numbers of output and input vertices.

Property 5 ([6, Lemma B.9]) Any labelled open graph $\Gamma = (G, I, O, \lambda)$ with focussed Pauli flow has $2^{|O|-|I|}$ distinct focussed sets.

Finally, Simmons gave an efficient algorithm that identifies independent generators for that group.

Property 6 ([6, Lemma B.10]) Given a labelled open graph Γ with focussed Pauli flow, there exists an algorithm that identifies $|O| - |I|$ independent generators for the group $\mathfrak{F}_\Gamma$. Furthermore, this algorithm completes in time polynomial in the number of vertices in Γ .

$\mathfrak{F}_\Gamma$ as a vector space

In [7, 19-20], Mitosek and Backens give an alternative interpretation for the collection of focussed sets $\mathfrak{F}_\Gamma$, by providing an isomorphism with the kernel of a ‘flow-demand matrix’ M_Γ introduced earlier and agnostic to the existence of a Pauli flow. That shows that $\mathfrak{F}_\Gamma$ also forms a vector space.

Definition 33 (Flow-demand matrix, [7, Definition 3.4]) Let $\Gamma = (G, I, O, \lambda)$ be a labelled open graph. The *flow-demand matrix* M_Γ is the $(n - n_O) \times (n - n_I)$ matrix with rows corresponding to non-outputs $\bar{O}$ and columns corresponding to non-inputs $\bar{I}$, where the row $M_{v,*}$ corresponding to the vertex $v \in \bar{O}$ satisfies the following for any $w \in \bar{I} \setminus \{v\}$:

- if $\lambda(v) \in \{X, XY\}$, then $M_{v,v} = 0$ and $M_{v,w} = \text{Adj}_{v,w}$,
- if $\lambda(v) \in \{Z, YZ, XZ\}$, then $M_{v,v} = 1$ and $M_{v,w} = 0$, and
- if $\lambda(v) \in \{Y\}$, then $M_{v,v} = 1$ (provided that the column exists) and $M_{v,w} = \text{Adj}_{v,w}$.

Theorem 4 ([7, Theorem 3.23]) Let $\Gamma = (G, I, O, \lambda)$ be a labelled open graph and let M_Γ be its flow-demand matrix. Then $\mathfrak{F}_\Gamma \simeq \text{Ker} M_\Gamma$.

Remark 7 In the proof of the theorem it is shown that such an isomorphism is given by the indicator function $\mathbf{1}_\bullet^{\bar{I}}$ and its inverse is given by the support $\text{supp}(\bullet)$.

Provided that Γ having a focussed Pauli flow implies $\text{rank} M_\Gamma = |V| - |O|$ and with the rank theorem, it is straightforward through that approach to verify Property 5.

Mitosek and Backens also define the ‘order-demand matrix’ N_Γ which will be used later in a proof.

Definition 34 (Order-demand matrix, [7, Definition 3.5]) Let $\Gamma = (G, I, O, \lambda)$ be a labelled open graph. The *order-demand matrix* N_Γ is the $(n - n_O) \times (n - n_I)$ matrix with rows corresponding to non-outputs $\bar{O}$ and columns corresponding to non-inputs $\bar{I}$, where the row $N_{v,*}$ corresponding to the vertex $v \in \bar{O}$ satisfies the following for any $w \in \bar{I}$:

- if $\lambda(v) \in \{X, Y, Z\}$, then $N_{v,*} = 0$,
- if $\lambda(v) = YZ$, then $N_{v,v} = 0$ and $N_{v,w} = \text{Adj}_{v,w}$,

- if $\lambda(v) = XZ$, then $N_{v,v} = 1$ and $N_{v,w} = \text{Adj}_{v,w}$, and
- if $\lambda(v) = XY$, then $N_{v,v} = 1$ (provided that the v column exists) and $N_{v,w} = 0$.

3 Contributions

In the remainder of this report, we investigate focussed sets. While they can be described as sets of vertices with trivial net effect when used for corrections and their collection has both a group structure and a vector space structure, what are their interpretation? What can they be used for?

3.1 Towards new properties

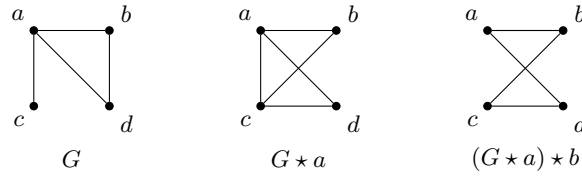
During the *Parcours Recherche*, some time was spent studying the effects of graph operations such as local complementation on focussed sets.

Definition 35 Let $G = (V, E)$ be a graph and $u \in V$ a vertex. The *local complementation of G about the vertex u* is the operation resulting in the graph

$$G \star u := \{V, E \Delta \{(v, w) \text{ such that } (v, u), (w, u) \in E \text{ and } v \neq w\}\}.$$

From [14, p. 12], the definition says that $G \star u$ is a graph that has the same vertices as G , but in which two neighbors v and w of u are connected if and only if they are not connected in G , and all other edges are the same as in G .

Example 21 [13, Example 5.2.2] Here is a graph G with four vertices a, b, c, d , its local complementation about a , $G \star a$, and the local complementation of $G \star a$ around b , $(G \star a) \star b$.



The following result gives conditions under which a focussed set remains one after applying a local complementation to the graph and consequently relabelling the vertices. Note that we solely focus on planar measurements, ie. $\lambda(w) \in \{XY, XZ, YZ\}$.

Lemma 5 Let $\Gamma = (G, I, O, \lambda)$ be a labelled open graph, $\mathcal{A} \subseteq \bar{I}$ a focussed set of Γ , that is $\mathcal{A} \in \mathfrak{F}_\Gamma$. Denoting $\Gamma' = (G \star u, I, O, \lambda')$ where $u \in \bar{O} \cap \bar{I}$ and λ' is defined from λ and u as in [14, Lemma 3.1], we have that $\mathcal{A}$ is no longer a focussed set of Γ' , that is $\mathcal{A} \notin \mathfrak{F}_{\Gamma'}$, if and only if one of the following conditions hold:

- $u \in \mathcal{A}$

- ii. $u \in \text{Odd}_G(\mathcal{A}) \setminus (N_G(u) \setminus \mathcal{A}) \wedge \lambda(u) = XZ$
- iii. $u \notin \text{Odd}_G(\mathcal{A}) \wedge (\exists w \in \bar{O} \cap N_G(u) \cap \mathcal{A} \cap \overline{\text{Odd}_G(\mathcal{A})} \setminus \{u\})$
- iv. $u \in \text{Odd}_G(\mathcal{A}) \wedge (\exists w \in \bar{O} \cap N_G(u) \cap \bar{\mathcal{A}} \cap \overline{\text{Odd}_G(\mathcal{A})}, \lambda(w) = XY)$

Proof The used definition of focussed sets requires us to verify three conditions:

(FX) Let $w \in \bar{O} \cap \mathcal{A}$. Since $\mathcal{A} \in \mathfrak{F}_\Gamma$, we have that $\lambda(w) \in \{XY, X, Y\}$, ie. $\lambda(w) = XY$ in our restrictive case. From the definition of λ' in [14, Lemma 3.1], we have that:

- if $w = u$, then $\lambda'(w) = XZ$ and (FX) can not be satisfied. This proves i.
- otherwise, $\lambda'(w) = XY$ and (FX) is satisfied

Finally, (FX) is verified if and only if $u \notin \bar{O} \cap \mathcal{A}$, ie. $u \notin \mathcal{A}$.

(FZ) Let $w \in \bar{O} \cap \text{Odd}_{G \star u}(\mathcal{A})$. From [13, Lemma 8.3.1] we have that:

$$\bar{O} \cap \text{Odd}_{G \star u}(\mathcal{A}) = \bar{O} \cap (\text{Odd}_G(\mathcal{A}) \Delta M) \text{ where } M = \begin{cases} N_G(u) \cap \mathcal{A} & \text{if } u \notin \text{Odd}_G(\mathcal{A}), \\ N_G(u) \setminus \mathcal{A} & \text{if } u \in \text{Odd}_G(\mathcal{A}) \end{cases}$$

Using the definition of Δ and the distributivity of intersection over union, we obtain:

$$\begin{aligned} \bar{O} \cap (\text{Odd}_G(\mathcal{A}) \Delta M) &= \bar{O} \cap ((\text{Odd}_G(\mathcal{A}) \setminus M) \cup (M \setminus \text{Odd}_G(\mathcal{A}))) \\ &= (\bar{O} \cap (\text{Odd}_G(\mathcal{A}) \setminus M)) \cup (\bar{O} \cap (M \setminus \text{Odd}_G(\mathcal{A}))) \\ &= (\bar{O} \cap \text{Odd}_G(\mathcal{A}) \cap \bar{M}) \cup (\bar{O} \cap M \cap \overline{\text{Odd}_G(\mathcal{A})}) \end{aligned}$$

Let's pursue by considering two distinct cases:

- $w \in \bar{O} \cap \text{Odd}_G(\mathcal{A}) \cap \bar{M} \subset \bar{O} \cap \text{Odd}_G(\mathcal{A})$: In this case and since $\mathcal{A} \in \mathfrak{F}_\Gamma$, we have that $\lambda(w) \in \{XZ, YZ\}$.
 - * $w = u$: Let's consider the value of $\lambda(u)$.
 - $\lambda(u) = XZ$: Then $\lambda'(u) = XY$ and (FZ) will not be satisfied (ii).
 - $\lambda(u) = YZ$: Then $\lambda'(u) = YZ$ and (FZ) is satisfied.
 - * $w \neq u$: Because we have $w \notin N_G(u)$, $\lambda'(w) = \lambda(w)$ and (FZ) is satisfied.
- $w \in \bar{O} \cap M \cap \text{Odd}_G(\mathcal{A})$:
 - * $u \notin \text{Odd}_G(\mathcal{A})$: In this case, $M = N_G(u) \cap \mathcal{A}$, so $w \in \mathcal{A}$ and since $\mathcal{A} \in \mathfrak{F}_\Gamma$ we have $\lambda(w) = XY$. If $w = u$, then $\lambda'(w) = XZ$ and (FZ) is satisfied, otherwise $\lambda'(w) = XY$ and it is not, proving iii.
 - * $u \in \text{Odd}_G(\mathcal{A})$: Since $w \in \text{Odd}_G(\mathcal{A})$, we can say that $w \neq u$. And because $w \in N_G(u)$, we can say that $\lambda'(w) = XY$ if $\lambda(w) = XY$ (iv) and $\lambda'(w) \in \{XZ, YZ\}$ if $\lambda(w) \in \{XZ, YZ\}$.

(FY) This last condition is trivially satisfied in our case (planar measurements only).

□

3.2 Adding focussed sets to correction sets

The rest of the *Parcours Recherche* was spent working on the following questions: What happens when one adds (symmetric difference) focussed sets to correction sets of Pauli flows ? Can focussed sets help us converge towards some Pauli flow optimality criteria ? Are some bases of the vector space $\mathfrak{F}_\Gamma$ "better" than others ?

3.2.1 Preservation of extensiveness

We start by adding one focussed set to one correction set of a given extensive correction function, and see whether it preserves its extensiveness.

Lemma 6 Let Γ be a labelled open graph and let c be a focused extensive correction function on Γ . Also let $\mathcal{A} \subseteq \mathfrak{F}_\Gamma$ and $v \in \bar{O}$. Define the correction function c' as follows:

$$c'(u) := \begin{cases} c(u) \Delta \mathcal{A} & \text{if } u = v, \\ c(u) & \text{otherwise,} \end{cases}$$

and for $u \in \bar{O}$ let the following set:

$$T_{\lambda(u)}(\mathcal{A}, c(v)) := \begin{cases} \text{Odd}(\mathcal{A}) \setminus \text{Odd}(c(v)) & \text{if } \lambda(u) = YZ, \\ \mathcal{A} \setminus c(v) & \text{if } \lambda(u) = XY, \\ \text{Odd}[\mathcal{A}] \setminus \text{Odd}[c(v)] & \text{if } \lambda(u) = XZ. \end{cases}$$

Then, $\prec_C$ being the transitive closure of the induced relation $\triangleleft_C$, we have that:

$$c' \text{ is focused extensive} \iff \forall u \in \bar{O} \cap T_{\lambda(u)}(\mathcal{A}, c(v)), \neg(u \prec_C v)$$

Proof The correction matrix C' associated with c' can be expressed as follows:

$$C'_{u,w} = \begin{cases} C_{u,w} \oplus \mathbb{1}_{u \in \mathcal{A}} & \text{if } w = v, \\ C_{u,w} & \text{otherwise} \end{cases}$$

From [7, Theorem 3.1] c' is focused extensive if, and only if, the following two facts hold:

- (i.) $M_\Gamma C' = \text{Id}_{\bar{O}}$
- (ii.) $N_\Gamma C'$ is the adjacency matrix of a DAG

As of now we will show that the last two facts hold if, and only if, the condition stated in the lemma is satisfied.

- (i.) This fact holds as soon as c is focused extensive. Indeed, C' can be written as $C + F$, where F is a matrix whose column all belong to $\text{Ker}(M_\Gamma)$ ¹⁰. As a consequence,

$$M_\Gamma C' = M_\Gamma C + M_\Gamma F = \text{Id}_{\bar{O}} + 0 = \text{Id}_{\bar{O}}$$

- (ii.) Let us denote D (resp. D') the directed graph associated with $N_\Gamma C$ (resp. $N_\Gamma C'$). Because c is focused extensive, D is a DAG and we are wondering whether D' is still one too.

From D to D' , we are only adding some $u \rightarrow v$ edges and removing some other $w \rightarrow v$ edges. That is, we are only affecting incoming edges of v . As a consequence, we only need to worry about a new $u \rightarrow v$ creating a cycle in D' .

¹⁰More precisely, F is the matrix whose column are all null except the v -th which is $\mathbb{1}_{\mathcal{A}}$.

However, using [7, Lemma 3.10], we can write the following equivalences:

$$\begin{aligned}
\text{Adding } u \rightarrow v &\iff (N_\Gamma C)_{u,v} = 0 \text{ and } (N_\Gamma C')_{u,v} = 1 \\
&\iff \begin{cases} \lambda(u) = YZ, & u \notin \text{Odd}(c(v)) \text{ and } u \in \text{Odd}(c'(v)) \\ \lambda(u) = XY, & u \notin c(v) \text{ and } u \in c'(v) \\ \lambda(u) = XZ, & u \notin \text{Odd}[c(v)] \text{ and } u \in \text{Odd}[c'(v)] \end{cases} \\
&\iff \begin{cases} \lambda(u) = YZ, & u \notin \text{Odd}(c(v)) \text{ and } u \in \text{Odd}(c(v)) \Delta \text{Odd}(\mathcal{A}) \\ \lambda(u) = XY, & u \notin c(v) \text{ and } u \in c(v) \Delta \mathcal{A} \\ \lambda(u) = XZ, & u \notin \text{Odd}[c(v)] \text{ and } u \in \text{Odd}[c(v)] \Delta \text{Odd}[\mathcal{A}] \end{cases} \\
&\iff u \in T_{\lambda(u)}(\mathcal{A}, c(v)) \text{ by definition of } \Delta
\end{aligned}$$

Furthermore, adding $u \rightarrow v$ creates a cycle in D' if and only if u is reachable from v in D . However, [7, Lemma 3.12] helps us show the following equivalences:

$$\begin{aligned}
u \text{ is reachable from } v \text{ in } D &\iff \exists s_1, \dots, s_n, \quad v \rightarrow s_1 \rightarrow \dots \rightarrow s_n \rightarrow u \\
&\iff \exists s_1, \dots, s_n, \quad (N_\Gamma C)_{v,s_1} = 1, \dots, (N_\Gamma C)_{s_n,u} = 1 \\
&\iff \exists s_1, \dots, s_n, \quad s_1 \triangleleft_C v, \dots, u \triangleleft_C s_n \\
&\iff u \prec_C v
\end{aligned}$$

The result follows by contraposition. $\square$

Because $\mathfrak{F}_\Gamma$ is closed under the associative symmetric difference operator Δ , adding multiple focussed sets to one correction set is equivalent to adding one big focussed set to one correction set.

Lemma 7 Let Γ be a labelled open graph and let c be a focused extensive correction function on Γ . Also let $\mathcal{A}_1, \dots, \mathcal{A}_k$ be k focussed sets of Γ and $v \in \bar{O}$. Define the correction function c' as follows:

$$c'(u) := \begin{cases} (c(u) \Delta \mathcal{A}_1) \Delta \dots \Delta \mathcal{A}_k & \text{if } u = v, \\ c(u) & \text{otherwise,} \end{cases}$$

That is, we are successively adding the focussed sets $\mathcal{A}_1, \dots, \mathcal{A}_k$ to $c(v)$. Then, $\prec_C$ being the transitive closure of the induced relation $\triangleleft_C$, we have that:

$$c' \text{ is focused extensive} \iff \forall u \in \bar{O} \cap T_{\lambda(u)}(\Delta_{i \in \{1, \dots, k\}} \mathcal{A}_i, c(v)), \neg(u \prec_C v)$$

The general case where we are adding multiple focussed sets to multiple correction sets is more complicated to study, but we can give the following result:

Lemma 8 Let Γ be a labelled open graph and let c be a focused extensive correction function on Γ . Also let $(\mathcal{A}_v)_{v \in \bar{O}}$ be a family of focussed sets of Γ . Define the correction function c' as follows:

$$\forall v \in \bar{O}, \quad c'(v) = c(v) \Delta \mathcal{A}_v$$

If $D = (V, A)$ (resp. $D' = (V', A')$) denotes the directed graph associated with $N_\Gamma C$ (resp. $N_\Gamma C'$), we have $V' = V$ and

$$A' = A \cup \{u \rightarrow v, u \in T_{\lambda(u)}(\mathcal{A}, c(v)), v \in \bar{O}\} \setminus \{u \rightarrow v, u \in \bar{T}_{\lambda(u)}(\mathcal{A}, c(v)), v \in \bar{O}\}$$

That is, we are adding $u \rightarrow v$ arcs for $u \in T_{\lambda(u)}(\mathcal{A}, c(v))$ and removing $u \rightarrow v$ arcs for $u \in \bar{T}_{\lambda(u)}(\mathcal{A}, c(v))$, where $T_{\lambda(u)}(\mathcal{A}, c(v))$ is defined upper and

$$\bar{T}_{\lambda(u)}(\mathcal{A}, c(v)) = \begin{cases} \text{Odd}(\mathcal{A}) \cap \text{Odd}(c(v)) & \text{if } \lambda(u) = YZ, \\ \mathcal{A} \cap c(v) & \text{if } \lambda(u) = XY, \\ \text{Odd}[\mathcal{A}] \cap \text{Odd}[c(v)] & \text{if } \lambda(u) = XZ. \end{cases}$$

Remark 8 The result from Lemma 6 might be interesting from an algorithm point of view. Instead of having to check for new cycles with a full depth/breadth-first search from v , we now just need to check reachability from v for $u \in T_{\lambda(u)}(\mathcal{A}, c(v))$. For large graphs and small focussed sets where $|T_{\lambda(u)}(\mathcal{A}, c(v))| \ll |V|$, this might be a significant improvement. However, note that it requires storing $\prec_C$.

3.2.2 Effect on depth

Back to adding only one focussed set to one correction set of a given focused extensive correction function, we also wondered what happens to the depth, a notion we define further. In that goal, we first need a few definitions.

Definition 36 A *partially ordered set* (or *poset*) is a pair $P = (X, \leq)$ consisting of a set X and a partial order $\leq$ on X .

Property 7 If c is a focused extensive correction function, then $(\bar{O}, \prec_c)$ is a poset.

Remark 9 Strictly speaking, the standard definition of a poset requires the order relation to be reflexive, which $\prec_c$ is not since it is a *strict* partial order. However, $\prec_c$ can be easily extended to a non-strict partial order by defining $u \preceq_c v := (u = v \text{ or } u \prec_c v)$. This distinction is anyway not significant here, so we will refer to $(\bar{O}, \prec_c)$ as a poset by abuse.

Definition 37 Let $P = (X, \leq)$ be a poset.

- We define *chains* of P as subsets of X that are totally ordered for $\leq$, ie. S is a chain of P if

$$\forall a, b \in S, \quad a \leq b \text{ or } b \leq a.$$

- Similarly, we define *antichains* of P as subsets of X in which no two distinct elements are comparable, ie. $\tilde{S}$ is an antichain of P if

$$\forall a, b \in \tilde{S} \text{ with } a \neq b, \quad \neg(a \leq b \text{ or } b \leq a).$$

Definition 38 We say that a chain (resp. antichain) S *starts from* a (resp. *ends in* a) if $a \in S$ and

$$\forall b \in S \setminus \{a\}, \quad a \leq b \quad (\text{resp. } b \leq a),$$

that is, if a is minimal (resp. maximal).

Definition 39 The *length* of a chain S is $|S| - 1$, ie. the size of the corresponding subset minus one.

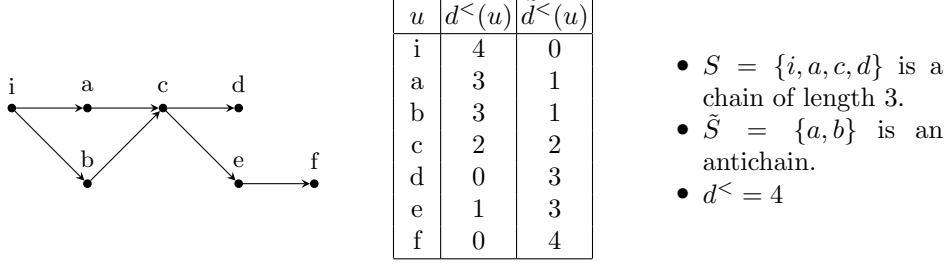


Fig. 2 A poset and its associated function values and properties.

Definition 40 For $P = (X, \leq)$ a poset and $u \in X$, we define $d^<(u)$, the *depth* of u , as the length of the longest chain of P starting from u .

We also define $\tilde{d}^<(u)$, which will refer to the length of the longest chain of P ending in u .

Definition 41 We finally define $d^<$, the *depth* of P , as the quantity $\max_{u \in X} d(u)$. What is sometimes called the *height* of P in the literature is then $d^< + 1$.

Definition 42 The *depth* $d^<$ of a focused Pauli flow $(c, \prec)$ is the depth of the poset $(\bar{O}, \prec)$.

Remark 10 The latest definition is coherent with [21, Definition 4]. Indeed, $(V_k^<)_{k=0 \dots d^<}$ is by construction a partition of $V(G) = \bar{O}$ into a minimum number of antichains, $d^< + 1$. By Mirsky's theorem, the height of the associated poset equals that quantity, and to our definition the depth of a poset equals its height minus one. Moreover, we even have that $(V_k^<)_{k=0 \dots d^<} = (\{u \in \bar{O}, d^<(u) = k\})_{k=0 \dots d^<}$.

Definition 43 The *length* of an antichain $\tilde{S}$ is $|\tilde{S}|$, ie. the size of the corresponding subset.

Definition 44 For P a poset, we also define its *width* $w(P)$ as the length of its longest antichain.

Remark 11 The width of a poset P equals the length of the longest element of any partition of P into antichains.

Definition 45 The *width* $w^<$ of a focused Pauli flow $(c, \prec)$ is the width of the poset $(\bar{O}, \prec)$. To the latest remarks, that is equivalent to say that:

$$w^< := \max_{0 \leq k \leq d^<} |\{u \in \bar{O}, d^<(u) = k\}| = \max_{0 \leq k \leq d^<} |V_k^<|$$

Definition 46 ([21, adapted from Definition 5]) Let Γ be a labelled open graph and let $(c, \prec)$ and $(c', \prec')$ be two Pauli flows of Γ . We say that $(c, \prec)$ is *more delayed* than $(c', \prec')$ if

$$\forall k, |\{u \in \bar{O}, d^<(u) \leq k\}| \geq |\{u \in \bar{O}, d^{\prec'}(u) \leq k\}|$$

and there exists a k such that the inequality is strict.

Definition 47 ([21, adapted from Definition 5]) A Pauli flow is said to be *maximally delayed* if there exists no Pauli flow of the same labelled open graph that is more delayed.

Property 8 The depth of a maximally delayed focussed Pauli flow is *optimal*, in the sense that no other Pauli flow of the same labelled open graph has a smaller depth.

Proof Let $(c, \prec)$ be a maximally delayed focussed Pauli flow and $(c', \prec')$ be another Pauli flow. $(c, \prec)$ being maximally delayed, we have that:

$$\forall k, |\{u \in \bar{O}, d^\prec(u) \leq k\}| \geq |\{u \in \bar{O}, d^{\prec'}(u) \leq k\}|$$

In particular,

$$|\{u \in \bar{O}, d^\prec(u) \leq d^{\prec'}(u)\}| \geq |\{u \in \bar{O}, d^{\prec'}(u) \leq d^\prec(u)\}|.$$

But recall that $d^\leq = \max_{u \in X} d^\leq(u)$; it follows that the RHS is exactly $|\bar{O}|$, so that the LHS is also $|\bar{O}|$ and we have $\forall u \in \bar{O}, d^\prec(u) \leq d^{\prec'}(u)$. Finally, $d^\prec \leq d^{\prec'}$ which is what we wanted. $\square$

With all that material, we can finally give conditions under which adding one focussed set to one correction set will not increase the depth.

Lemma 9 Let Γ be a labelled open graph and let c be a maximally delayed focused extensive correction function on Γ . Also define c' as in Lemma 6 with $\mathcal{A} \subseteq \mathfrak{F}_\Gamma$ and $v \in \bar{O}$ satisfying the conditions, so that c' is focused extensive. Then, we have that:

$$d^{\prec c} = d^{\prec' c} \iff \forall u \in T_{\lambda(u)}(\mathcal{A}, c(v)), \quad d^{\prec c}(u) \leq d^{\prec c} - \tilde{d}^{\prec c}(v) - 1$$

Proof The flow $(c, \prec_c)$ being maximally delayed, $d^{\prec c}$ is optimal and $d^{\prec' c} \geq d^{\prec c}$. As a consequence, we only need to study the conditions under which the depth might increase between c and c' .

And such conditions can be expressed as follows: *we are adding $v \prec_c u$ for some $u \in \bar{O}$ and there existed $\tilde{S}_v$ of length y_1 ending in v and S_u of length y_2 starting in u s.t. $y_1 + 1 + y_2 > d^{\prec c}$ (we are creating a new longest chain).*

From the previous Lemma, we can characterize the $u \in \bar{O}$ s.t. we are adding $v \prec_c u$; they are the elements of $T_{\lambda(u)}(\mathcal{A}, c(v))$. By contraposing the latest condition and $\tilde{d}^{\prec c}(v)$ being defined as the length of the longest chain ending in v , we can conclude. $\square$

The effects on the width remain an open question, as there appears to be no explicit condition ruling its increase or decrease.

4 Future work

In the remaining part, we discuss ideas for future work and questions that were raised during the *Parcours Recherche*.

Graphical representations

In this report, the ZX-calculus was used for introducing MBQC but not for our discussions around focussed sets. Indeed, in the ZX representation of a computation in the one-way model, one can highlight the qubits which belong to a focussed set, however there seems to be no visual interpretation of focussed sets in the ZX-calculus.

Graph operations

We studied the effect of local complementation on $\mathfrak{F}_\Gamma$ in Section 3.1. The result could be simplified if a simple relationship between M_Γ and $M_{\Gamma'}$ - where $\Gamma = (G, I, O, \lambda)$ and $\Gamma' = (G \star u, I, O, \lambda)$ - was found. Moreover, there could exist more interesting insights between local complementation - or other operations such as pivoting [14] - and $\mathfrak{F}_\Gamma$.

Adding focussed sets to correction sets

Why would one add focussed sets to correction sets? The report gives a few results on the effects of such an operation, but the objective behind it must be further understood. Given a focussed Pauli flow - for example, the one obtained by the algorithm from [7] - what are the properties of those which can be reached by adding focussed sets to correction sets?

Bases for $\mathfrak{F}_\Gamma$

As a vector space over $\mathbb{F}_2$, $\mathfrak{F}_\Gamma$ admits many bases. Among them, are some "better" than others? For that question to be answered, one would have to define to characterize the quality of a base. This would probably be linked with the flows they help reach.

Depth and width of a flow

While the depth of a flow had already been defined in the literature, it does not seem to be the case for the width. These two notions are probably subject to many discussions, since they imply different things on the physical implementation of the computation. For example, the depth of a flow was described as an upper bound on the depth of the corresponding deterministic one-way quantum computation [21] - that is, the number of sequential measurement 'steps' it requires. On the other hand, the width of a flow might be viewed as the maximal number of simultaneous measurements required for each measurement 'step'.

Appendix A Python work

Throughout this *Parcours Recherche*, the Python programming language and Jupyter notebooks were extensively used to manipulate the mathematical objects under study. In November 2024, Piotr Mitosek - co-author of [7] - provided access to a private GitHub repository containing an implementation of the Pauli flow algorithm from the paper. This resource was particularly helpful for visualizing Pauli flows and partial orders, automating the computation of flow and order-demand matrices, and extracting bases of focussed sets, among other tasks.

To better suit the needs of the project, the repository was forked and several improvements were made to the code:

- A **CorrectionFunction** class was created, with a matrix representing the function as an attribute and useful methods such as **isFocussedExtensive** (based on [7, Theorem 3.23]) and **getInducedRelation**.
- A **getT** function was implemented to return the $T_{\lambda(u)}(\mathcal{A}, c(v))$ sets defined earlier, and used to verify newly proposed criteria.
- A **getDepth** method was added to the existing **PauliFlow** class.
- An option was introduced to allow users to choose whether to print all focussed sets or just a basis.
- Finally, larger labelled open graphs with Pauli flow were required, which was tedious to construct manually. This motivated the following open question: How can one automatically generate labelled open graphs of a given size that admit Pauli flow?

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